Open-World Explained-Rarity Filtering for Health-Insurance Claim Fraud: Problem Formulation, Benchmark Design and Preliminary Synthetic Evidence

Ishant Pant
Lead Author
Department of Data Science
Manipal University Jaipur
Dehmi Kalan, Jaipur, Rajasthan 303007, India
ishantpant2332@gmail.com

Dr. Sandeep Gupta
Research Supervisor
Department of Data Science
Manipal University Jaipur
Dehmi Kalan, Jaipur, Rajasthan 303007, India

***Abstract*—Health-insurance fraud detectors trained on known fraud categories can force novel schemes into familiar classes and confuse clinically rare but legitimate claims with fraud. We formulate *explained-rarity filtering*: a claim is escalated only when it is unfamiliar and unexplained by clinical and provider evidence. We report tabular and temporal relation-summary pilots, standalone and controlled deep-model comparisons, and a custom typed-node temporal graph benchmark with a matched relation-free control. In 12 synthetic folds, the graph arm achieved known-class accuracy 0.957 ± 0.011 and macro-F1 0.652 ± 0.095, but novel-mechanism AUROC was 0.696 ± 0.166 and recall at nominal $\alpha = 0.05$ was only 0.108 ± 0.126. The relation-free control performed slightly better on mean AUROC (0.706 ± 0.190) and novel recall (0.127 ± 0.127), so this experiment does not demonstrate a benefit from graph message passing. Collusion AUROC remained near/below chance. The graph encoder is custom PyTorch R-GCN/HGT-style message passing, not canonical PyG HGTConv/RGCNConv, and uses synthetic Admission/Diagnosis proxies. No experiment uses real claim-level fraud labels.**

***Index Terms*—health-insurance fraud, open-world recognition, conformal prediction, anomaly detection, graph neural networks**

I. Introduction

Large claim volumes in health-insurance systems, including IRDAI-regulated insurance and the AB-PMJAY scheme in India, make exhaustive manual review impractical. Fraud mechanisms such as upcoding, unbundling, phantom billing and collusive referral rings also change over time. Three problems follow.

1) **Closed-world classification.** A softmax classifier assigns a new scheme to the nearest known class, often with high confidence.
2) **Rarity is not fraud.** Pure anomaly scores flag legitimate severe cases because they are rare, complex and expensive.
3) **Alert fatigue.** Raw scores carry no statistical meaning, so a fixed threshold drifts and review budgets are exceeded.

Contributions. (1) A problem formulation, *explained-rarity filtering*, that separates unfamiliarity from unexplainedness. (2) A pipeline (OW-ERF) combining post-hoc novelty fusion, split-conformal rejection and a heteroscedastic clinical expectation filter. (3) A benchmark protocol with a synthetic claim generator, leave-one-mechanism-out evaluation and leakage control. (4) Preliminary results from tabular, deep-model, temporal relation-summary and typed-node temporal graph experiments, including mechanism-specific failure cases.

II. Related Work

Open-set and open-world recognition. Open-set recognition [1], [2] and out-of-distribution (OOD) baselines [3]–[5] score unfamiliarity from logits or embeddings. On graphs, OpenWGL [6] rejects unseen-class nodes with a variational graph autoencoder, and GNNSafe [7] applies energy-based detection to graph neural networks.

Graph fraud and anomaly detection. CARE-GNN [8] counters camouflage with reinforcement-learned neighbor selection, and BWGNN [9] addresses heterophily. DOMINANT [10] and the PyGOD library [11] cover reconstruction-based detection, while BOND [12] and GADBench [13] provide benchmarks. These methods output novelty or anomaly scores and do not separate rare-but-clinically-explained claims from unexplained ones.

Conformal prediction. Split-conformal methods [14]–[16] give finite-sample marginal guarantees under exchangeability. Barber et al. [17] study relaxations, and Huang et al. [18] apply conformal prediction to graph neural networks. Temporal claim streams violate exchangeability, so our guarantee is approximate and is tested empirically.

Heteroscedastic regression. Predicting a mean and a variance under a Gaussian likelihood [19], [20] lets the expected spread grow with severity.

Healthcare fraud. Prior work applies supervised and unsupervised learning to Medicare-style data [21].

III. Problem Formulation

Let a claim v have features x_v, a graph neighborhood, and a label y_v from benign, known mechanisms, or novel mechanisms. Training sees only benign and known mechanisms. At test time a claim is *unfamiliar* if its conformal p-value satisfies $p_v \le \alpha$, and *unexplained* if its clinical inconsistency ratio a_v exceeds a calibrated threshold τ. A claim is escalated if and only if it is both.

To prevent temporal leakage, the embedding of claim v at adjudication time t uses only the causal neighborhood

$$\mathcal{N}_r(v;t) = \{u : (u,v) \in E_r,\ t(u,v) \le t\}. \tag{1}$$

IV. Method

Fig. 1 shows the pipeline.

A. Encoder

The target architecture specifies a timestamped heterogeneous graph of claims, beneficiaries, hospitals, doctors, admissions, packages and diagnoses, with time-aware edges and an R-GCN or HGT encoder using PyTorch Geometric [22]–[24]; the earlier relation-summary pilot did not implement this arbitrary-node PyG encoder. A separate dependency-light typed-node temporal message-passing encoder has since been evaluated over 12 synthetic folds; those measured results are reported in the typed-node temporal graph benchmark subsection below. It is a custom PyTorch R-GCN/HGT-style implementation rather than the canonical PyG HGTConv/RGCNConv. Three complementary branches are separately implemented and trained in the released synthetic benchmark: an LSTM over recent claim-history sequences, a 1D-CNN over ordered procedure-code tokens, and a Transformer over relational-context tokens. The standalone experiment compares each architecture independently rather than claiming that it is already fused with a full HGT/R-GCN. The three models use different input modalities by design, so the comparison is a comparison of architecture-plus-modality pipelines. A separate relation-summary pilot implements causal relation-specific attention and additional OW-ERF stages, but it is not a full arbitrary-node heterogeneous graph encoder.

B. Novelty score

From classifier logits z, the energy is

$$E(x) = -T \log \sum_k \exp(z_k/T). \tag{2}$$

The Mahalanobis score $M(h)$ is the minimum over known classes of the squared Mahalanobis distance of embedding h to the class mean under the class covariance. With F_E and F_M the empirical CDFs of E and M on training data, the fused score is

$$u = \tfrac{1}{2}F_E(E) + \tfrac{1}{2}F_M(M). \tag{3}$$

C. Conformal rejection

On a calibration set disjoint from training and containing only known-class claims, with n calibration scores u_i, the p-value is

$$p_v = \frac{1 + \#\{i : u_i \ge u_v\}}{n+1}, \tag{4}$$

and a claim is rejected as unfamiliar when $p_v \le \alpha$ with $\alpha = 0.05$.

D. Explained-Rarity Filter

A heteroscedastic model trained on benign claims only predicts a mean μ_j and standard deviation σ_j for $j \in \{\text{log-amount}, \text{length of stay}\}$ from *exogenous* inputs (a severity proxy from prior history, comorbidity count, age) by minimizing

$$\mathcal{L}_g = \frac{1}{N}\sum_i \sum_j \Big[\ln \sigma_{ij} + \frac{(o_{ij}-\mu_{ij})^2}{2\sigma_{ij}^2}\Big]. \tag{5}$$

The inconsistency ratio is

$$a_v = \max_j \max\Big\{0, \frac{o_j - \mu_j}{\sigma_j}\Big\}. \tag{6}$$

The threshold τ is set as a quantile of a_v on benign calibration claims rather than by hand. The expectation model must not condition on fields a fraudster controls, such as the claimed diagnosis code; otherwise upcoding would make fraud look explained.

E. Additional OW-ERF pilot components

The separate relation-summary pilot implements episodic pseudo-novel energy-margin training, energy/Mahalanobis fusion, split-conformal rejection, legitimate-only clinical expectation and structural-reference thresholds, explained-rarity filtering, HDBSCAN grouping (with DBSCAN fallback when HDBSCAN is unavailable), cluster cards and audit-oriented metrics. It uses past-only entity summaries over six relation types, not a PyG HGT/R-GCN over arbitrary typed nodes. A distinct custom typed-node temporal encoder is now available in `owerf_temporal_heterograph.py`; its metrics are intentionally not folded into this pilot's results. The standalone LSTM/CNN/Transformer experiment is a separate benchmark and must not be reported as the full graph model.

V. Experimental Protocol

Data. A synthetic generator produces benign claims, 5% rare-severe legitimate claims, and four mechanisms: upcoding, unbundling, phantom billing, and collusion (represented in the prototype by two proxy features only). The fraud rate is 10%.

Splits and metrics. Three experiments are reported separately. The earlier tabular prototype uses five random seeds and its own i.i.d. split. The standalone deep comparison uses

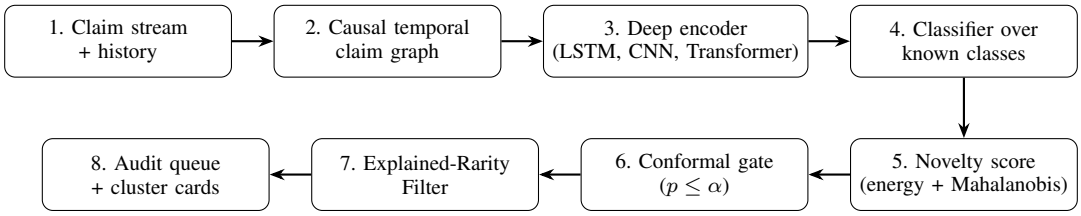


Fig. 1. Pipeline of OW-ERF. Stages 1–4 learn representations; stages 5–8 make the decision.

TABLE I
LEAVE-ONE-MECHANISM-OUT RESULTS (SYNTHETIC DATA, TABULAR PROTOTYPE, MEAN OF 5 SEEDS). RECALL IS ON THE HELD-OUT MECHANISM; "RARE FLAG" IS THE FRACTION OF RARE-SEVERE LEGITIMATE CLAIMS FLAGGED.

Held-out mechanism	Recall conf.	Recall +ERF	Rare flag conf.	Rare flag +ERF
Upcoding	0.574	0.560	0.026	0.006
Unbundling	0.493	0.372	0.020	0.003
Phantom billing	0.496	0.411	0.133	0.026
Collusion (proxy)	0.586	0.211	0.016	0.002

three generator seeds (42, 43, 44), four leave-one-mechanism-out folds, an internal known-only validation subset, a separate known-only calibration split, and an untouched test split, yielding 36 independent model fits. Known-class accuracy is calculated only on test claims from classes observed during training; novel-mechanism AUROC/AP treat the held-out mechanism as positive for novelty scoring. The v0.2 relation-summary pilot uses chronological train, selection, calibration and deployment windows. These results are synthetic, and the standalone benchmark does not establish real-world performance.

Baseline. Isolation Forest [25], with its alert count matched to the number of OW-ERF escalations.

VI. PRELIMINARY RESULTS

Table I retains the earlier tabular-only reference results (means over five seeds). The conformal stage rejects between 4.3% and 4.6% of benign claims, close to the nominal 5% on this i.i.d. synthetic split; this does not demonstrate validity under temporal drift. After the filter, benign escalation falls to about 0.5%. The filter cuts flags on rare-severe legitimate claims most clearly for phantom billing ($13.3\% \rightarrow 2.6\%$). It also lowers recall: for collusion the drop is large ($0.586 \rightarrow 0.211$), because those claims look clinically plausible and would require graph structure that this prototype lacks. At a matched alert count, Isolation Forest recalls only 0.02–0.04 of the three attribute-level mechanisms (0.20 for collusion) while flagging 22%–28% of rare-severe legitimate claims. All of these results are conditional on the generator.

A. Standalone deep-model comparison

Table II reports the mean and standard deviation across the 12 model/fold/seed runs per architecture (four held-out mechanisms by three generator seeds). Known-class accuracy excludes the held-out mechanism; novelty metrics treat the held-out mechanism as novel. This distinction is important because the model may classify known classes reasonably while failing to reject unfamiliar fraud.

The Transformer has the strongest mean known-class accuracy and novel AUROC in this 5-epoch pilot, but the high dispersion and low mean novel recall show that performance is not yet robust. The average known false-rejection rates at $\alpha = 0.05$ were 0.050 (LSTM), 0.047 (1D-CNN), and 0.045 (Transformer); this calibration target applies to known classes, not unseen fraud. The full JSON/CSV and selected checkpoints are generated by the dedicated GitHub Actions workflow. These metrics are synthetic generator results, not real insurance claim results.

B. Controlled architecture comparison with shared inputs

The native standalone comparison above changes both architecture and input modality. To isolate the architecture effect more fairly, we ran a paired controlled comparison: the LSTM, a continuous-input 1D-CNN, and Transformer received the same standardized six-token, 16-channel relational-context tensor and the same padding mask. Scaling was fitted only on valid training tokens; padded positions were zeroed, the LSTM packed valid sequence lengths, the CNN used mask-aware pooling, and the Transformer used its key-padding mask. The experiment used the same four held-out mechanisms and generator seeds 42, 43, and 44, for 36 model fits per mode (72 total). Checkpoints were selected using validation cross-entropy; the test split was not used for model selection. The 25-epoch pilot is preserved in a separate report; Table III presents the completed 60-epoch follow-up.

The large LSTM and CNN known-class-accuracy changes under shared inputs show that the original score gap was substantially confounded by modality: collusion information is encoded in history, unbundling in code tokens, and upcoding/phantom cues in the claim-context view. The Transformer is numerically unchanged because its native input was already that context view. Under shared context, novelty AUROC improves for LSTM and CNN, but dispersion remains high and novel recall at the conformal threshold is still low/variable. The shared-context known false-rejection rates were 0.048 ± 0.019 (LSTM), 0.054 ± 0.015 (1D-CNN), and 0.061 ± 0.021 (Transformer); the Transformer remains slightly above the nominal 5% rate. Validation-selected best epochs reached the 60-epoch cap in 3/12 shared-context LSTM runs and 1/12 shared-context CNN runs (none of the Transformer runs),

TABLE II
INDEPENDENT DEEP-MODEL RESULTS ON THE SYNTHETIC HELD-OUT-MECHANISM BENCHMARK (MEAN ± STANDARD DEVIATION ACROSS 12 FOLDS/SEEDS PER MODEL).

Model	Known accuracy	Known macro-F1	Novel AUROC	Novel AP	Novel recall ($\alpha = 0.05$)
LSTM	0.630 ± 0.183	0.380 ± 0.105	0.576 ± 0.189	0.139 ± 0.277	0.156 ± 0.288
1D-CNN	0.565 ± 0.114	0.279 ± 0.048	0.561 ± 0.126	0.092 ± 0.093	0.120 ± 0.159
Transformer	0.809 ± 0.115	0.646 ± 0.137	0.693 ± 0.142	0.142 ± 0.215	0.156 ± 0.251

TABLE III
CONTROLLED DEEP-MODEL COMPARISON ON SYNTHETIC DATA (60-EPOCH MAXIMUM; MEAN ± STANDARD DEVIATION ACROSS 12 FOLD/SEED RUNS PER MODEL). NATIVE MODE USES EACH MODEL'S INTENDED INPUT; SHARED-CONTEXT MODE GIVES ALL MODELS THE SAME STANDARDIZED TOKENS.

Model	Native acc.	Shared acc.	Native AUROC	Shared AUROC	Shared AP	Shared novel recall ($\alpha = 0.05$)
LSTM	0.519 ± 0.098	0.979 ± 0.010	0.539 ± 0.130	0.721 ± 0.210	0.169 ± 0.105	0.250 ± 0.166
1D-CNN	0.896 ± 0.016	0.973 ± 0.016	0.546 ± 0.136	0.759 ± 0.171	0.188 ± 0.141	0.354 ± 0.325
Transformer	0.983 ± 0.009	0.983 ± 0.009	0.816 ± 0.090	0.816 ± 0.090	0.174 ± 0.103	0.349 ± 0.228

so some LSTM/CNN folds may still benefit from a larger epoch budget or early-stopping study. With three generator seeds, these descriptive metrics do not establish a statistically decisive architecture ranking. All results are synthetic and do not establish performance on real insurance claims. The exact JSON/CSV reports and selected checkpoints are committed in the controlled-comparison results directory; the 25-epoch pilot remains separately archived.

C. Typed-node temporal graph benchmark

The separate custom temporal heterograph encoder was evaluated in a four-mechanism, three-seed synthetic benchmark (12 runs; 2,400 generated claims per run; eight-epoch cap; batch size 64). The chronological split used 55% for training, 10% for model selection, 15% for known-only calibration, and 20% for deployment. Input scaling used the training window only. Each query block read entity summaries aggregated from claims strictly earlier than the block; claims in a query block did not update the entity state seen by other claims in that same block. The custom PyTorch model uses typed Claim, Beneficiary, Hospital, Doctor, Provider, Admission, Package and Diagnosis nodes, with eight relation-specific historical message channels (provider, hospital, doctor, beneficiary, admission, package, diagnosis and referral target). Provider and referral-target relations share a Provider node namespace but use distinct relation-specific projections in this custom model. It is not the canonical PyG HGTConv/RGCNConv implementation. Admission is a one-per-claim synthetic proxy and Diagnosis is a severity-bin proxy.

Across all 12 folds, known-class accuracy was 0.957 ± 0.011 and macro-F1 was 0.652 ± 0.095. Novel-mechanism AUROC averaged 0.696 ± 0.166, average precision 0.090 ± 0.069, and recall at the nominal $\alpha = 0.05$ conformal rejection threshold 0.108 ± 0.126. The known false-rejection rate was 0.039 ± 0.013 and the explained-rarity escalation rate was 0.040 ± 0.013. These are descriptive synthetic results; a nominal conformal level does not guarantee coverage under temporal drift.

Table IV shows substantial mechanism dependence. Phantom-billing AUROC was 0.892 ± 0.019, while upcoding AUROC was 0.751 ± 0.016. Unbundling novel recall was zero across the three seeds. Collusion AUROC was 0.455 ± 0.040 and recall was 0.026 ± 0.044, so provider and referral-target channels did not solve collusion detection in this generator. The relation-free control had higher mean AUROC in the upcoding, unbundling and phantom-billing folds and comparable below-chance collusion AUROC. The matched experiment therefore gives no evidence that this graph branch improves novelty detection on the current generator. The explained-rarity gate's novel recall equaled raw conformal novel recall in all 12 full-graph runs. Although explicit rare-severe denominators exist in the current per-fold reports, zero false rejection applies only to this generated cohort and is not evidence of real-world safety.

The tracked full-graph summary is under `results/temporal_heterograph_full/`; per-fold reports and checkpoints are in GitHub Actions artifacts for [the full-graph run](https://github.com/ishantpant2121/MUJ-DS-2430010379/actions/runs/40680366943). The matched no-graph control is recorded in the same workflow artifact, with aggregate comparison in `docs/TEMPORAL_GRAPH_BENCHMARK_RESULTS.md`. These findings should not be conflated with provider-level `PotentialFraud` labels and do not validate performance on real insurance claims.

D. Matched relation-free temporal graph control

To test whether entity message passing adds value beyond claim features, we ran a matched relation-free control. The same custom encoder, train/selection/calibration/deployment windows, parameter shapes, seed initialization, loss and downstream novelty/gating pipeline were retained, but all entity-message contributions were set to zero. Each arm used the same four held-out mechanisms and three generator seeds, yielding 12 fits per arm. Table V reports mean ± standard deviation across those folds/seeds.

TABLE IV
MECHANISM-SPECIFIC RESULTS FOR THE TYPED-NODE TEMPORAL GRAPH BASELINE (SYNTHETIC DATA; MEAN ± STANDARD DEVIATION ACROSS THREE GENERATOR SEEDS PER HELD-OUT MECHANISM).

Held-out mechanism	Novel AUROC	Novel AP	Novel recall ($\alpha = 0.05$)
Upcoding	0.751 ± 0.016	0.131 ± 0.071	0.150 ± 0.066
Unbundling	0.688 ± 0.022	0.036 ± 0.008	0.000 ± 0.000
Phantom billing	0.892 ± 0.019	0.161 ± 0.030	0.255 ± 0.138
Collusion	0.455 ± 0.040	0.031 ± 0.019	0.026 ± 0.044

TABLE V
MATCHED TEMPORAL GRAPH AND NO-GRAPH CONTROL ON SYNTHETIC DATA (MEAN ± STANDARD DEVIATION, 12 RUNS PER ARM).

Metric	Full graph	No-graph control
Known-class accuracy	0.957 ± 0.011	0.959 ± 0.011
Known macro-F1	0.652 ± 0.095	0.666 ± 0.100
Novel AUROC	0.696 ± 0.166	0.706 ± 0.190
Novel average precision	0.090 ± 0.069	0.098 ± 0.074
Novel recall at $\alpha = 0.05$	0.108 ± 0.126	0.127 ± 0.127
Known false-rejection rate	0.039 ± 0.013	0.038 ± 0.012
Gated novel recall	0.108 ± 0.126	0.121 ± 0.125
Escalation rate	0.040 ± 0.013	0.040 ± 0.012

The relation-free control is slightly better on average for the main novelty metrics. This is a negative ablation result for the current graph design and simulator: the experiment does not demonstrate that graph message passing improves detection. Because the two arms were evaluated on simulated data and only three generator seeds, the comparison is descriptive and does not prove that graph information is useless in real claims. Both arms show poor, variable novel recall; the experiment must be reported transparently rather than selecting the better-looking arm. The detailed paired comparison and artifact links are in the repository's temporal graph benchmark results document.

VII. LIMITATIONS

All measured experiments reported here use parametric synthetic data, not real claims or claim-level fraud labels from an insurer. Known-class accuracy and novel-mechanism AUROC are distinct tasks and must not be collapsed into one accuracy claim. The standalone LSTM/CNN/Transformer branches use generated proxies and are not fused end-to-end with the temporal graph encoder. The typed-node graph result is a custom PyTorch R-GCN/HGT-style baseline, not the canonical PyG HGTConv/RGCNConv implementation. Admission is represented by one synthetic node per claim, and Diagnosis is a severity-bin proxy rather than an ICD-10 diagnosis node. The graph benchmark is only an initial 12-run evaluation (four held-out mechanisms by three generator seeds, with an eight-epoch cap). Novel recall at the nominal conformal threshold is low on average and varies by mechanism; collusion AUROC is near chance and unbundling recall is weak. The explained-rarity gate's novel recall matched raw conformal recall in all 12 folds, so this benchmark does not demonstrate incremental filtering of those novel cases. The aggregate report does not preserve the denominator for the rare-severe-benign subset; its reported zero false-rejection metric is therefore excluded from interpretation. The v0.2 relation-summary pilot uses prior-event summaries. Package rates remain parametric placeholders rather than a reviewed catalogue from the official HBP 2.2 annex, and these reported experiments do not ingest Synthea exports. Synthea itself does not supply claim-level fraud-mechanism labels by default. The conformal guarantee is marginal under exchangeability, while temporal streams may violate this assumption; the explained-rarity gate is outside that guarantee. Manipulable severity evidence may explain away suspicious claims. Broader baselines, ablations, shortcut audits, subgroup/fairness analyses, repeated confirmatory experiments, provenance-checked clinical/HBP reference data, and appropriately labelled real-claim validation remain necessary. Any future deployment must retain human review because incorrect escalation can delay or burden legitimate care.

VIII. CONCLUSION AND FUTURE WORK

The repository contains the policy-recommendation prototype, tabular and relation-summary OW-ERF pilots, standalone and controlled deep-model experiments, and a separate typed-node temporal graph baseline with a matched relation-free control. In the current 12-run synthetic benchmark, graph known accuracy was 0.957 ± 0.011 and novel AUROC was 0.696 ± 0.166, with novel recall at $\alpha = 0.05$ of only 0.108 ± 0.126. The relation-free arm slightly outperformed it on mean AUROC, AP and recall. Collusion AUROC remained near chance/below chance, and unbundling recall was zero in the full graph arm. The explained-rarity gate did not add novel recall over conformal rejection. These results support further methodological investigation, not deployable fraud detection. Next steps include investigating the negative ablation result and weak mechanisms, broadening baselines, auditing

shortcuts and subgroup effects, curating source-verified clinical/HBP references, and evaluating on appropriately labelled real claims with human review.

Data and Code Availability

No patient data are used. The synthetic generator, tabular prototype, standalone deep-model benchmark, temporal relation-summary pilot, typed-node temporal graph encoder and evaluation script, Synthea graph adapter, and policy-recommendation prototype are available at https://github.com/ishantpant2121/MUJ-DS-2430010379.
The standalone comparison is reproducible from `scripts/train_evaluate_deep_models.py`; it writes JSON/CSV metrics and 36 model checkpoints to `results/standalone_deep_models/`. Aggregate graph-baseline results are stored under `results/temporal_heterograph_full/`; the full 12-fold run's predictions and model checkpoints are retained in the GitHub Actions artifact. All deep and graph-style results use synthetic data; the separate public provider benchmark uses provider-level `PotentialFraud` labels and is not claim-level mechanism validation.

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